

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and deployed production **GenAI systems** using Python and LLM integrations, delivering measurable business impact by automating key enterprise workflows.
- Built automated **ML pipelines** encompassing data preparation, model training, evaluation, and production deployment, integrating with Apple's enterprise data platform infrastructure.
- Developed **LLM-based solutions** for enterprise use cases, applying prompt engineering techniques and RAG patterns to enhance information retrieval and task automation.
- Collaborated with cross-functional teams and business stakeholders to scope, develop, and ship **AI and ML systems** that addressed real business problems and improved operational efficiency.
- Engineered **Python services** on AWS Lambda for real-time data processing, handling approximately 2,800 events per second during peak loads and replacing a legacy batch system.
- Refactored relational database queries and implemented strategic indexing for the enterprise data warehouse, reducing analytical report execution times by **31%** across core datasets.

SKILLS

ML Engineering – Machine Learning, Generative AI, LLMs, RAG, Prompt Engineering, Agentic Frameworks, Model Deployment, ML Pipelines, MLOps, Anomaly Detection, Forecasting

Programming Languages – Python, Java, SQL, R

Data Platforms & Tools – Enterprise Data Warehouse, Distributed Data Systems, Large-Scale Data Processing, Power BI, Tableau, Power Query, DAX, Data Modelling, Data Warehousing

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes

Backend Development – Spring Boot, REST APIs, FastAPI, Flask

DevOps & CI/CD – Jenkins, Git, Code Reviews

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

Enterprise GenAI Assistant Platform

- Developed a **Generative AI assistant** leveraging RAG for internal knowledge base queries, improving information retrieval speed by **40%** for the support team.
- Engineered Python data pipelines to ingest and process terabytes of enterprise documentation, creating embeddings for efficient semantic search within the assistant.
- Integrated the assistant with existing enterprise systems via **REST APIs**, enabling natural language interaction for common operational tasks.

Predictive Maintenance ML Model

- Built and trained a **LightGBM classifier** on sensor data from enterprise equipment (1.5M records), achieving **0.92 AUC** for predicting potential failures.
- Developed feature engineering scripts in Python (pandas, NumPy) to extract time-series and statistical features relevant to equipment health.
- Deployed the trained model as a **FastAPI service** on AWS EC2, providing real-time anomaly scoring with p95 latency under **200ms**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)